Teacher Wu Kuan | Must-Miss Exam Question — Conditional Probability with Two Dice

English lecture notes · YouTube Short r-WX1cShzxg

1. The trap question

Given at least one die shows 6, what is the probability both show 6?

2. The method

- “Given …” shrinks the sample space to only the outcomes that fit the condition.
- 6×6 grid: at least one 6 = $6 + 6 - 1 = 11$ cells (or $36 - 5 \times 5 = 11$).
- Both 6 = 1 cell → P = 1/11 (not 1/36, not 1/6).

3. Parent check method

Have your child shade the “given” cells first and count them before counting the favorable ones.

Practice

1. Two dice: how many outcomes have at least one 6?
2. Given at least one die is 6, find P(both are 6).
3. Given the sum is 8, find P(both dice are even).
4. Given the FIRST die is 6, find P(both are 6).
5. Given at least one die shows 1, find P(sum is 7).

Answers

1. 11
2. 1/11
3. 3/5 (sum 8: 5 outcomes; both even: (2,6), (4,4), (6,2))
4. 1/6
5. 2/11 ((1,6) and (6,1) out of 11)

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